Artificial Intelligence

Supervised Learning

Evaluation: Confusion Matrix

- It is an N x N matrix used for evaluating the performance of a classification model
  - where N is the number of target classes.
  - The matrix compares the actual target values with those predicted by the machine learning model.

For a binary classification problem, we would have a 2 x 2 matrix as shown in figure with 4 values:

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

- The target/true variable has two values: **Positive** or **Negative**
- The **columns** represent the **actual values** of the target variable
- The **rows** represent the **predicted values** of the target variable

Evaluation: Confusion Matrix

- TP – True Positive
  - Number of correct positive predictions that are actually positive
- FP – False Positive
  - Number of incorrect positive predictions that are actually negative
- TN – True Negative
  - Number of correct negative predictions that are actually negative
- FN – False Negative
  - Number of incorrect negative predictions that are actually positive

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Example

- Predict how many people are infected with a contagious virus in times before they show the symptoms, and isolate them from the healthy population;
  - The two values for our target will be: Sick and Not Sick
  - Dataset: 1000 data samples:
  - Train Binary Classifier: Neural Network
  - Network Prediction:

The total outcome values are:

	Actual	
Predicted	TP=30	FP=30
	FN=10	TN=930

ID	Actual Sick?	Predicted Sick?	Outcome
1	1	1	TP
2	0	0	TN
3	0	0	TN
4	1	1	TP
5	0	0	TN
6	0	0	TN
7	1	0	FP
8	0	1	FN
9	0	0	TN
10	1	0	FP
:	:	:	:
1000	0	0	FN

Evaluation Metric

- Accuracy:

It can be defined as how much data is correctly classified

$$\text{Accuracy} = \frac{TP \quad +TN}{TP+FN+FP+TN}$$

$$\text{Accuracy} = \frac{30+930}{30+10+30+930}$$

$$\text{Accuracy} = 0.96 \;\Rightarrow 96\%$$

Evaluation Metric: Accuracy

- Network Prediction: 96% Accuracy
- It tells: “I can predict sick people 96% of the time”.
- However, it is doing the opposite.
- It is predicting the people who will **not get sick** with 96% accuracy while the sick are spreading the virus! Because of TN

For Imbalanced Dataset: Classification Accuracy is not a correct measure

Evaluation Metric

- Precision:
  - It tells how many of the correctly predicted cases actually turned out to be positive. OR
  - It can be defined as how much returned data is correct

$$\text{Precision} = \frac{TP}{TP+FP}$$

- Recall:
  - It tells how many of the actual positive cases we were able to predict correctly with our model OR
  - It can be defined as how much correct data is returned

$$\text{Recall} = \frac{TP}{TP+FN}$$

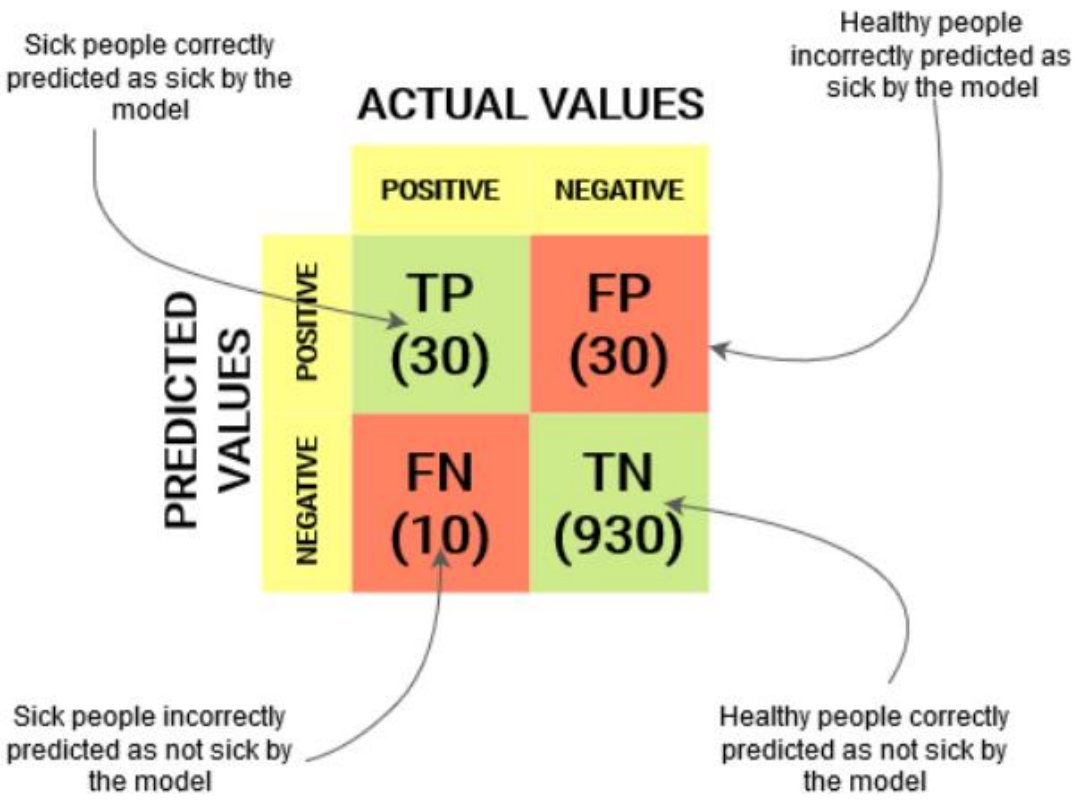
Evaluation Metric

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Precision} = \frac{30}{30+30} = 0.5$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$\text{Recall} = \frac{30}{30+10} = 0.75$$



50% percent of the correctly predicted cases turned out to be positive cases. Whereas 75% of the positives were successfully predicted by model.

Analysis: Precision vs Recall

- Precision is a useful metric in cases where False Positive is a higher concern than False Negatives.

Precision is important in music or video recommendation systems, e-commerce websites, etc. Wrong results could lead to customer churn and be harmful to the business.

- Recall is a useful metric in cases where False Negative trumps False Positive.

Recall is important in medical cases where it doesn't matter if we raise a false alarm, but the actual positive cases should not go undetected!

- In previous example; Recall would be a better metric because we don't want to accidentally discharge an infected person and let them mix with the healthy population thereby spreading the contagious virus.

Evaluation Metric:

- F1-Score
  - In practice, when we try to increase the precision of our model, the recall goes down, and vice-versa. The F1-score captures both the trends in a single value:

F1-score is a harmonic mean of Precision and Recall, and so it gives a combined idea about these two metrics. It is maximum when Precision is equal to Recall.

$$F1 - \text{Score} = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}} = 0.6$$

Confusion Matrix: Multiclass

- IRIS Dataset
  - The dataset has 3 flowers as outputs or classes, Versicolor, Virginia, Setosa.

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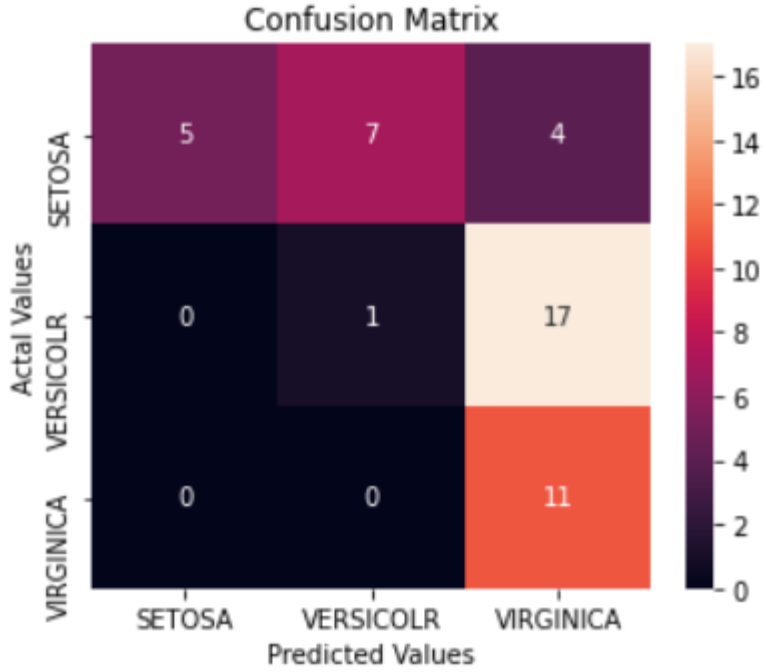
  - How to calculate TP, FP, TN, FN

Confusion Matrix: Multiclass

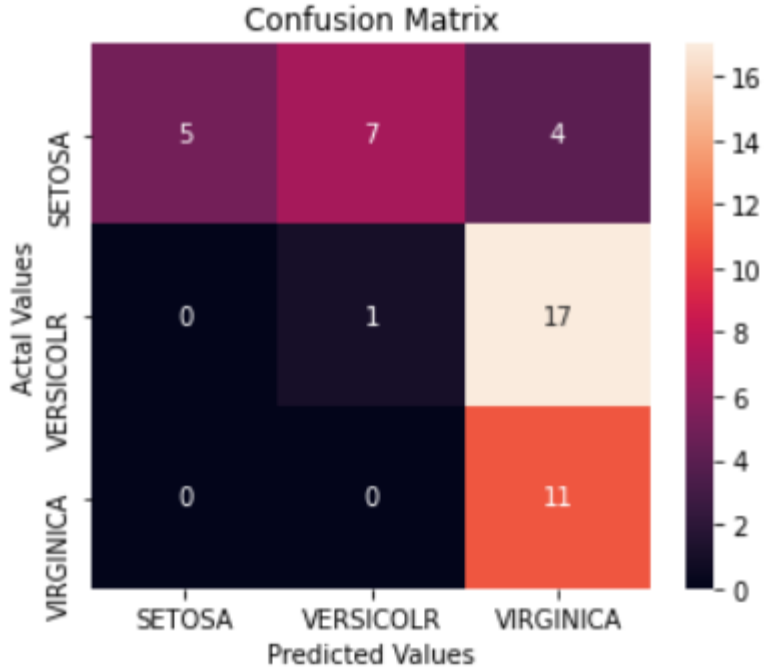
- **How to calculate FN, FP, TN, TP :**
- FN: The False-negative value for a class will be the sum of values of corresponding rows except for the TP value.



For SETOSA:
FN= 7 + 4

Confusion Matrix: Multiclass

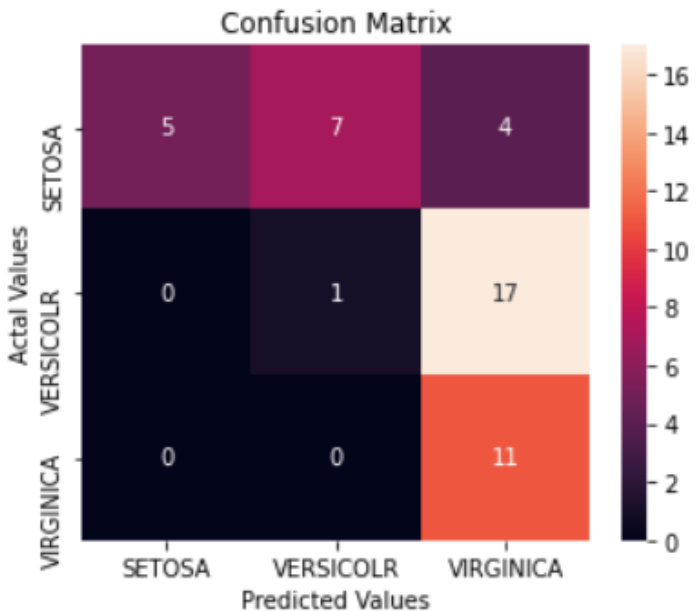
- **How to calculate FN, FP, TN, TP :**
- FP: The False-positive value for a class will be the sum of values of the corresponding column except for the TP value.



For SETOSA:
FN= 7 + 4 (R1C2+ R1C3)
FP= 0 + 0 (R2C1 + R3C1)

Confusion Matrix: Multiclass

- **How to calculate FN, FP, TN, TP :**
- TN: The True Negative value for a class will be the sum of values of all columns and rows except the values of that class that we are calculating the values for.



For SETOSA:

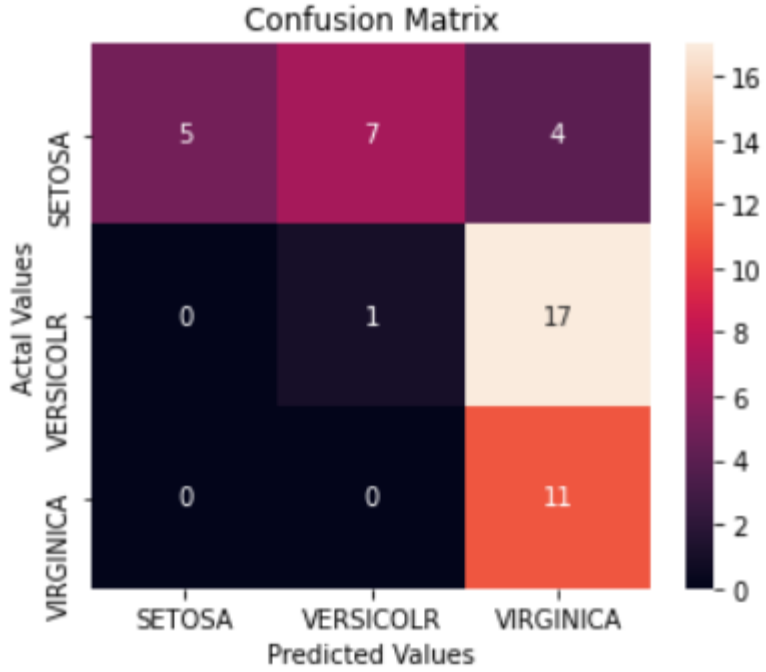
FN= 7 + 4 (R1C2+ R1C3)

FP= 0 + 0 (R2C1 + R3C1)

TN= 1 + 17 + 0 + 11 (R2C2 + R2C3 + R3C2 + R3C3

Confusion Matrix: Multiclass

- **How to calculate FN, FP, TN, TP :**
- TP: The True positive value is where the actual value and predicted value are the same.



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FN= 7 + 4 (R1C2+ R1C3)
FP= 0 + 0 (R2C1 + R3C1)
TN= 1 + 17 + 0 + 11 (R2C2 + R2C3 + R3C2 + R3C3

TP= 5 (R1C1)

Confusion Matrix: Multiclass

- **How to calculate FN, FP, TN, TP :**
- FN: The False-negative value for a class will be the sum of values of corresponding rows except for the TP value.
- FP: The False-positive value for a class will be the sum of values of the corresponding column except for the TP value.
- TN: The True Negative value for a class will be the sum of values of all columns and rows except the values of that class that we are calculating the values for.
- TP: The True positive value is where the actual value and predicted value are the same.

Multiclass Classification: Confusion Matrix

- MNIST Dataset: labels[0-9]

```
[[5825    1   37    9   12   20   31   13   12   22]
 [   0 6657   53   23   14   34   25   46  101   17]
 [   1   19 5627   44   12   13    6   52   12    3]
 [   4   21   50 5827    0   67    2   24  111   59]
 [   6   12   48    6 5480   33    8   40   33   61]
 [  10    7    9   79    1 5095   38    2   13    8]
 [  26    1   30   14   43   45 5789    4   20    2]
 [   1    4   29   34    6    1    0 5872    3   26]
 [  38   12   63   51   10   51   19   11 5447   24]
 [  12    8   12   44  264   62    0  201   99 5727]]
```

Macro-average Precision

system output \ gold labels	urgent	normal	spam	
urgent	8	10	1	$\text{precision}_u = \frac{8}{8+10+1}$
normal	5	60	50	$\text{precision}_n = \frac{60}{5+60+50}$
spam	3	30	200	$\text{precision}_s = \frac{200}{3+30+200}$
	$\text{recall}_u = \frac{8}{8+5+3}$	$\text{recall}_n = \frac{60}{10+60+30}$	$\text{recall}_s = \frac{200}{1+50+200}$	

Figure 6.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2), how many documents from c_1 were (in)correctly assigned to c_2

Class 1: Urgent

	true urgent	true not
system urgent	8	11
system not	8	340

$$\text{precision} = \frac{8}{8+11} = .42$$

Class 2: Normal

	true normal	true not
system normal	60	55
system not	40	212

$$\text{precision} = \frac{60}{60+55} = .52$$

Class 3: Spam

	true spam	true not
system spam	200	33
system not	51	83

$$\text{precision} = \frac{200}{200+33} = .86$$

$$\text{macroaverage precision} = \frac{.42+.52+.86}{3} = \mathbf{.60}$$

The **weighted-averaged** F1 score is calculated by taking the mean of all per-class F1 scores

Micro-average Precision

		gold labels: urgent	normal	spam	
system output	urgent	8	10	1	$\text{precision}_u = \frac{8}{8+10+1}$
	normal	5	60	50	$\text{precision}_n = \frac{60}{5+60+50}$
	spam	3	30	200	$\text{precision}_s = \frac{200}{3+30+200}$
		$\text{recall}_u = \frac{8}{8+5+3}$	$\text{recall}_n = \frac{60}{10+60+30}$	$\text{recall}_s = \frac{200}{1+50+200}$	

Figure 6.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2), how many documents from c_1 were (in)correctly assigned to c_2

Micro-avg precision= TP1+TP2+TP3/ TP1+TP2+TP3+FP1+FP2+FP3

TP1=8, TP2= 60, TP3=200

FP1= 10+1

FP2= 50+5

FP3= 30 + 3

Micro-avg Precision= 268/268+99=0.73

- A micro-average is dominated by the more frequent class (in this case spam)
  - as the counts are pooled
- The macro-average better reflects the statistics of the smaller classes,
  - is more appropriate when performance on all the classes is equally important.

Evaluation

- In general, if you are working with an imbalanced dataset where all classes are equally important, using the **macro** average would be a good choice as it treats all classes equally.

- If you have an imbalanced dataset but want to assign greater contribution to classes with more examples in the dataset, then the weighted average is preferred.
  - This is because, in weighted averaging, the contribution of each class to the F1 average is weighted by its size.